Parkinson’s Tremor Detection

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Outline

- This project aims to develop a smart wristband for detecting Parkinson’s disease symptoms, specifically hand tremors, using motion sensors. By integrating an accelerometer and gyroscope (e.g., MPU6050) with a microcontroller like the ESP32, the system will capture and analyze wrist movements. Signal processing techniques such as filtering and Fast Fourier Transform (FFT) will be used to identify tremors in the 4–6 Hz range, characteristic of Parkinson’s. A vibration motor will be included to provide haptic feedback—either as a cue to assist movement or as an alert when tremors exceed a threshold. Data can be sent via Bluetooth to a mobile app for real-time display and logging. The system will be optimized for power efficiency and designed to be low-cost, making it suitable for continuous monitoring and potential use in clinical or home settings.

Objective

- To design and develop a low-cost, wearable wristband that detects Parkinson’s disease-related tremors using motion sensors, processes the data in real time, and provides haptic feedback to assist users, enabling continuous monitoring and early detection of motor symptoms without the need for complex medical equipment.

Introduction

- Parkinson’s disease is a progressive neurological disorder that primarily affects motor functions, leading to symptoms such as resting tremors, bradykinesia, and muscle rigidity. Early detection and continuous monitoring of these symptoms are crucial for effective disease management and improving patients' quality of life. Traditional clinical evaluations are often limited to short visits and can miss variations in symptoms throughout the day. This project proposes a smart wristband that uses motion sensors—specifically an accelerometer and gyroscope—to detect hand tremors associated with Parkinson’s disease in real time. By applying signal processing techniques to analyze sensor data, the system can identify tremor patterns and alert the user or caregiver when necessary. A vibration motor is included to provide haptic feedback, which can be used to cue movement or notify the user during severe tremor episodes. The wristband aims to be a cost-effective, non-invasive, and portable solution for continuous symptom tracking and assistance, making it useful in both clinical and home settings.

1.	Machine Learning Strategies for Parkinson Tremor Classification Using Wearable Sensor Data	Jesus Paucar-Escalante, Matheus Alves da Silva, Bruno De Lima Sanches, Aurea Soriano-Vargas, Laura Silveira Moriyama, Esther Luna Colombini	2025	arXiv (2025)	Reviewed ML methodologies for classifying Parkinsonian tremors, evaluating various data acquisition, preprocessing, and feature selection techniques.	effi li Fore in dis an
2.	Wearable-based Parkinson's Disease Severity Monitoring using Deep Learning	Jann Goschenhofer, Franz MJ Pfister, Kamer Ali Yuksel, Bernd Bischl, Urban Fietzek, Janek Thomas	2019	arXiv (2019)	Developed deep learning models for detecting motor states from wearable sensor data, employing transfer learning and class-weighting techniques.	Den deep can dete PD high con mor

Implementation

Hardware Design:

- **Components**: Use an ESP32 or Arduino microcontroller, accelerometer, gyroscope, vibration motor, and a Bluetooth module.
- **Sensor Placement**: Place the accelerometer and gyroscope on the wrist to measure hand movements and tremors.
- **Vibration Motor**: Provides haptic feedback for tremor-related cues or alerts.

Data Acquisition and Signal Processing:

- **Data Collection**: Capture motion data (accelerometer and gyroscope) at a high frequency (50-100 Hz).
- **Preprocessing**: Apply filters to remove noise and normalize data.
- **Feature Extraction**: Use metrics like RMS, ZCR, variance, and FFT to analyze tremor patterns.

Machine Learning Model:

- **Model Selection**: Train a classifier (e.g., SVM, Random Forest, CNN) to distinguish Parkinson's tremors from normal movements.
- **Training**: Use labeled tremor data to train and evaluate the model, optimizing for accuracy.
- **Evaluation**: Test the model using metrics like accuracy, precision, and recall.

Vibration Motor Integration:

- **Cueing & Tremor Suppression**: Provide vibration feedback to assist in movement (e.g., cue slow movements or suppress tremors).
- **Alert System**: Trigger a strong vibration if tremor severity exceeds a threshold.

Data Transmission and User Interface:

- **Mobile App**: Sync the wristband with a mobile app via Bluetooth to display real-time tremor data.
- **Cloud Storage**: Optionally store data in the cloud for long-term monitoring and analysis.

Power Management:

- Use sleep modes and low-power sensors to optimize battery life.
- Dynamically adjust the sampling rate based on detected movement.

Testing and Validation:

- Validate the device by comparing tremor data with clinical tools (e.g., UPDRS).
- Test on patients and controls to refine the system's accuracy and usability.

References

Bremm, R. P., Werle, A., Auer, C., Hertel, F., Gonçalves, J., & Koch, K. P. (2021). **TreCap: A wearable device to measure and assess tremor data of visually guided hand movements in real time**. *arXiv 2021*.

- URL: https://arxiv.org/abs/2107.04572
- Summary: This paper introduces a wearable device designed to assess hand tremors in real-time using tremor data.

Paucar-Escalante, J., Alves da Silva, M., De Lima Sanches, B., Soriano-Vargas, A., Silveira Moriyama, L., & Luna Colombini, E. (2025). **Machine learning strategies for Parkinson tremor classification using wearable sensor data**. *arXiv 2025*.

- URL: https://arxiv.org/abs/2501.03968
- Summary: This research reviews various machine learning strategies for classifying Parkinsonian tremors using sensor data.

Papadopoulos, A., Kyritsis, K., Klingelhoefer, L., Bostanjopoulou, S., Chaudhuri, K. R., & Delopoulos, A. (2020). **Detecting Parkinsonian tremor from IMU data collected in-the-wild using deep multiple-instance learning**. *arXiv 2020*.

- URL: https://arxiv.org/abs/2012.12232
- Summary: This paper presents a method for detecting Parkinsonian tremor using data from smartphone sensors in uncontrolled environments.

San-Segundo, R., Zhang, A., Cebulla, A., Panev, S., Tabor, G., Stebbins, K., Massa, R. E., Whitford, A., & De la Torre, F. (2020). **Parkinson's disease tremor detection in the wild using wearable accelerometers**. *MDPI Sensors 2020*.

- URL: https://www.mdpi.com/journal/sensors
- Summary: The paper demonstrates the use of wearable accelerometers for detecting Parkinson's tremors outside of clinical environments.

WHY?

Early Detection and Monitoring: Helps with early diagnosis and continuous tracking of disease progression.

Improved Patient Care: Provides real-time, objective data for personalized treatment and better symptom management.

Accessible and Non-Invasive: Offers a continuous, non-invasive way to monitor symptoms outside of clinical settings.

Empowerment and Quality of Life: Enhances patient mobility and independence through features like haptic feedback.

Real-Time Alerts and Remote Monitoring: Enables caregivers or family members to monitor symptoms remotely and intervene when necessary.